

AIRSENSE-AI

Anirudh S (Roll Number: 23035010660)

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B.Sc. (Honours) Data Science and Artificial Intelligence
Indian Institute of Technology Guwahati, India
anirudh.s@op.iitg.ac.in

ABSTRACT

Air pollution has become one of the most significant environmental challenges affecting public health and urban sustainability. Accurate estimation of the Air Quality Index (AQI) enables individuals and authorities to assess pollution levels and take appropriate preventive measures. This paper presents AirSense-AI, a machine learning-based web application developed for predicting AQI using six major atmospheric pollutants: PM_{2.5}, PM₁₀, CO, NO₂, SO₂, and O₃. A curated historical dataset containing air quality observations for four Indian metropolitan cities—Delhi, Mumbai, Chennai, and Kolkata—was preprocessed and used to train multiple regression models. Linear Regression, Random Forest, and XGBoost were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Among the evaluated models, Random Forest achieved the best performance with an RMSE of 7.91 and an R^2 score of 0.9936. The trained model was integrated into an interactive Streamlit dashboard that provides AQI prediction, historical city-wise comparisons, pollutant trend visualization, and health advisories. The developed system demonstrates how machine learning can support intuitive analysis and prediction of urban air quality through an accessible web interface.

Index Terms— Air Quality Index, Machine Learning, Random Forest, XGBoost, Streamlit, Data Visualization.

1. INTRODUCTION

Air pollution has emerged as one of the most critical environmental challenges worldwide, posing serious risks to public health, climate, and sustainable urban development. Rapid industrialization, increasing vehicular emissions, and population growth have significantly contributed to the deterioration of air quality in metropolitan regions. To quantify the level of pollution, the Air Quality Index (AQI) is widely used as a standardized indicator that reflects the combined impact of multiple atmospheric pollutants. Accurate estimation and analysis of AQI enable individuals and authorities to monitor pollution levels and adopt appropriate preventive measures.

Recent advancements in machine learning have provided efficient techniques for predicting AQI from pollutant concentrations. Compared to conventional statistical approaches, machine learning models are capable of capturing complex relationships among environmental variables, resulting in improved prediction accuracy. Furthermore, integrating predictive models with interactive visualization tools enhances accessibility and allows users to better understand historical air quality trends.

This paper presents **AirSense-AI**, a machine learning-based web application developed to predict AQI using six major atmospheric pollutants: PM_{2.5}, PM₁₀, CO, NO₂, SO₂, and O₃. The proposed system utilizes a curated historical air quality dataset containing observations from four Indian metropolitan cities: Delhi, Mumbai, Chennai, and Kolkata. Three regression algorithms, namely Linear Regression, Random Forest, and XGBoost, were trained and evaluated using standard performance metrics. Based on experimental evaluation, the Random Forest model achieved the best predictive performance and was integrated into the Streamlit application. In addition to AQI prediction, the application provides historical city-wise AQI comparison, pollutant trend visualization, and health advisory recommendations through an interactive dashboard.

The remainder of this paper is organized as follows. Section II presents the proposed methodology, including dataset preparation and machine learning workflow. Section III discusses the experimental results and application features. Finally, Section IV concludes the paper and outlines potential directions for future enhancements.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Air quality has become a major concern in urban environments due to increasing levels of atmospheric pollution caused by industrial activities, vehicular emissions, and rapid urbanization. Monitoring and predicting the Air Quality Index (AQI) is essential for assessing environmental conditions and promoting public health awareness. However, interpreting pollutant measurements individually can be difficult for non-technical users, making it challenging to understand overall air quality conditions. Therefore, there is a need for an intelligent and user-friendly system that can accurately predict AQI from pollutant concentrations while providing meaningful visualizations for historical air quality analysis.

2.2. Objectives

The primary objectives of this project are:

1. To preprocess and analyze a historical air quality dataset containing observations from four Indian metropolitan cities.
2. To develop machine learning models for predicting the Air Quality Index (AQI) using six major atmospheric pollutants.
3. To compare the performance of Linear Regression, Random Forest, and XGBoost regression models using standard evaluation metrics.

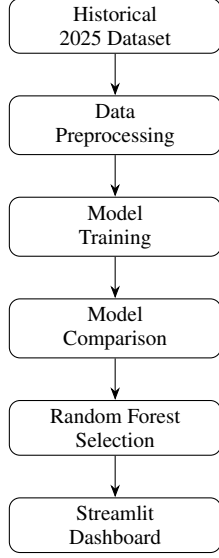


Fig. 1. Overall workflow of the proposed AirSense-AI system.

4. To select the most accurate prediction model and integrate it into a Streamlit-based web application.
5. To provide interactive dashboards for historical AQI comparison, pollutant trend visualization, and health advisory recommendations.

3. METHODOLOGY / APPROACH

The proposed AirSense-AI system follows a structured machine learning pipeline consisting of data preprocessing, model training, model evaluation, and deployment through an interactive Streamlit application. The overall workflow of the proposed system is illustrated in Figure 1.

3.1. Overall Workflow

The system begins with a curated historical air quality dataset containing observations from Delhi, Mumbai, Chennai, and Kolkata. After preprocessing and feature selection, multiple regression models are trained and evaluated. The best-performing model is deployed within a Streamlit application for AQI prediction and historical air quality analysis.

3.2. Dataset and Preprocessing

The dataset contains historical air quality measurements for four Indian metropolitan cities collected during 2025. Six atmospheric pollutants, namely PM2.5, PM10, CO, NO₂, SO₂, and O₃, are used as input features, while the Air Quality Index (AQI) serves as the prediction target. Data preprocessing involved handling missing values, formatting the dataset, and preparing numerical features for machine learning.

3.3. Machine Learning Models

Three regression algorithms were evaluated for AQI prediction:

- Linear Regression
- Random Forest Regressor

- XGBoost Regressor

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Based on the experimental results, the Random Forest model achieved the highest prediction accuracy and was selected for deployment within the AirSense-AI application.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

The proposed models were trained and evaluated using a historical air quality dataset containing observations from four Indian metropolitan cities: Delhi, Mumbai, Chennai, and Kolkata. Six atmospheric pollutants (PM2.5, PM10, CO, NO₂, SO₂, and O₃) were used as input features for predicting the Air Quality Index (AQI). The dataset was preprocessed to remove inconsistencies and prepare numerical features for model training.

Three regression algorithms, namely Linear Regression, Random Forest, and XGBoost, were implemented using the Scikit-learn and XGBoost libraries in Python. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). The Streamlit framework was used to develop the interactive web application for AQI prediction and historical air quality visualization.

4.2. Results

Table 1 summarizes the performance of the evaluated machine learning models.

Table 1. Performance Comparison of Regression Models

Model	MAE	RMSE	R^2
Linear Regression	22.42	29.78	0.9088
Random Forest	2.51	7.91	0.9936
XGBoost	3.34	9.70	0.9903

Among the evaluated models, the Random Forest Regressor achieved the best predictive performance with the lowest prediction error and the highest coefficient of determination. Consequently, it was selected as the final model for deployment within the AirSense-AI application.

Feature importance analysis indicated that PM2.5 and PM10 were the most influential predictors of AQI, followed by O₃, while CO, NO₂, and SO₂ contributed comparatively less to the prediction model. These observations highlight the dominant influence of particulate matter on urban air quality prediction.

The trained Random Forest model was integrated into a Streamlit-based dashboard that enables users to predict AQI by entering pollutant concentrations. The application also provides historical city-wise AQI comparison, pollutant trend visualization, and health advisory recommendations. Figure 2 presents the dashboard showing the average AQI comparison among the four selected cities, while Figure 3 illustrates the AQI prediction interface with sample pollutant values.

5. DISCUSSION

The experimental results demonstrate that ensemble learning methods outperform the baseline Linear Regression model for AQI pre-



Fig. 2. Average AQI comparison across Delhi, Mumbai, Chennai, and Kolkata from the AirSense-AI dashboard.

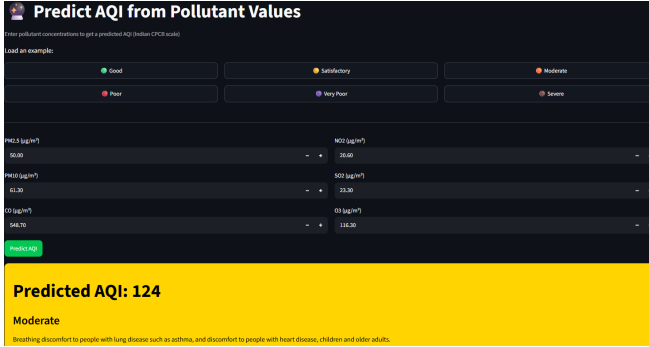


Fig. 3. AQI prediction interface of the AirSense-AI application using sample pollutant values.

diction. Among the evaluated algorithms, the Random Forest Regressor achieved the highest prediction accuracy with an RMSE of 7.91 and an R^2 score of 0.9936. Its superior performance can be attributed to its ability to model complex and non-linear relationships between atmospheric pollutants and the Air Quality Index. Although XGBoost also produced competitive results, Random Forest exhibited lower prediction error and was therefore selected as the final deployment model.

Feature importance analysis revealed that PM2.5 and PM10 were the dominant contributors to AQI prediction, indicating the significant influence of particulate matter on urban air quality. The comparatively lower importance of CO, NO₂, and SO₂ suggests that these pollutants had a smaller contribution within the selected dataset.

The developed Streamlit application successfully integrates the trained machine learning model with interactive visualizations, enabling users to predict AQI and analyze historical pollution trends across Delhi, Mumbai, Chennai, and Kolkata. The intuitive interface enhances accessibility by presenting technical information through graphical comparisons and health advisory recommendations.

Despite its promising performance, the proposed system has certain limitations. The model was trained using a historical dataset limited to four Indian metropolitan cities and one year of observations. Consequently, its predictions may not generalize to other geographical regions or future environmental conditions. Furthermore, the current implementation focuses on historical analysis and offline prediction without incorporating real-time air quality data or meteorological parameters.

Future improvements may include integrating live AQI data through public APIs, extending the dataset to additional cities and multiple years, incorporating weather variables, and exploring deep learning or time-series forecasting techniques for enhanced prediction accuracy.

6. CONCLUSION AND FUTURE WORK

This paper presented **AirSense-AI**, a machine learning-based web application for predicting and analyzing the Air Quality Index (AQI) using historical air quality data from four Indian metropolitan cities. The proposed system utilized six major atmospheric pollutants as input features and compared the performance of three regression algorithms: Linear Regression, Random Forest, and XGBoost. Experimental evaluation demonstrated that the Random Forest Regressor achieved the highest prediction accuracy with an RMSE of 7.91 and an R^2 score of 0.9936, making it the most suitable model for deployment.

The trained model was successfully integrated into a Streamlit-based dashboard that enables users to predict AQI, compare historical air quality across cities, visualize pollutant trends, and receive health advisory recommendations. The application provides an intuitive and interactive platform for understanding urban air quality, demonstrating the practical use of machine learning in environmental data analysis.

Future work may focus on incorporating real-time air quality data through public APIs, expanding the dataset to include additional cities and multiple years of observations, integrating meteorological variables such as temperature and humidity, and investigating advanced deep learning or time-series forecasting techniques to further improve prediction accuracy and system scalability.

7. ARTIFACTS AND DEMONSTRATIONS

The implementation and supporting resources for the proposed AirSense-AI system are made available through the following artifacts.

- **GitHub Repository:** <https://github.com/anlrudhks/AirSense-AI>
- **Project Demonstration (YouTube):** <https://youtu.be/bluYUN6dyTs>

The GitHub repository contains the complete source code, project structure, trained machine learning model, dataset preprocessing scripts, and the Streamlit application required to reproduce the presented results. The demonstration video provides an overview of the system architecture, model training process, application features, and a walkthrough of the AirSense-AI dashboard, including AQI prediction, historical city-wise comparison, pollutant trend visualization, and health advisory generation.

8. REFERENCES

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